

Luke Fox Williams, MIET

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MSc Mechatronics & Robotics graduate with hands-on experience in robotics, CAD, Embedded Systems, Programming, and Product Development. Proven ability to take projects from concept to functional prototype, including an adaptable in-pipe inspection robot. Academic Requirements for Chartered Engineer (CEng) status fully met through studies, & Registered member of the IET. Full & Clean Driver's License.

EDUCATION

University of Leeds

Sep. 2024 – Sep. 2025

MSc(Eng) Mechatronics & Robotics. Pass with Merit (Accredited by IET & IMechE)

- Conducted independent R&D of a vibration-based in-pipe inspection robot capable of adapting to pipes of varying diameter.
- Directed a 2-person team in theoretical design and development process for a proposed solution to the World Health Organisation topic of Hunger.
- Headed a team of 4 in developing a program for the Terasic DE1-SoC board utilising an LCD display during Embedded Systems.
 - Modules of Note & Skills Acquired:
 - **Embedded Systems/FPGA design for system on-chip** – Teambuilding, HDL/Verilog
 - **Control Systems Design** – Control Theory, MATLAB, Maths
 - **Power Electronics & Drives** – Technical, Analytical, Practical Engineering Skills
 - **Modern Industry Practice** – Teambuilding, Leadership, Organisation, R&D
 - **Programming** – Python, C++, MATLAB
 - **Professional Project** – SolidWorks, 3D Printing, R&D, Design Constraints

Staffordshire University

Sep. 2020 – July. 2024

BSc AI & Robotics. 2:1 with Hons

- Developed a computer vision algorithm capable of detecting an identification band in real-time.
- Thoroughly investigated and reported the ethical implications and considerations of AI in military, with specific regards to the Iron Dome defence system.
 - Modules of Note & Skills Acquired:
 - **Commercial Computing** – Tailored Design, Teambuilding, Presenting Information
 - **Robotic Programming & Vision** – C, C#, C++
 - **Robotic Modelling and Drone Skin Design** – Web Development, CAD
 - **Decision Analytics** – Excel, Data Analysis
 - **Robot Ethics in an AI World** – Research Documentation, Ethics Considerations
 - **Final Project** – Python, Computer Vision, AI Data Training, YOLO

PROJECTS

Adaptable in-pipe Vibration Robot

May 2025 – Aug. 2025

- Designed and built an adaptable in-pipe vibration robot using self-taught CAD and 3D printing techniques. Iteratively prototyped multiple designs, culminating in a functional working robot.
 - Additional technical detail and/or images available upon request.

PREVIOUS EMPLOYMENT

Self – Employed

Jan. 2020 – Aug. 2020

- Converted unpowered Bicycles into E-bikes during studies in Denmark. Demonstrated market analysis and sales skills.

SKILLS, TECHNOLOGIES & INTERESTS

- **Technical Skills:** Robotics, Rapid Prototyping, CAD, Embedded Systems, Computer Vision, Experimental Testing & Analysis, Programming
- **Business & Transferable Skills:** Research & Development, Product Development, Leadership, Teamwork & Communication, Market Research, Sales
- **Soft/Personal Skills:** Problem Solving, Critical Thinking, Team Collaboration, IT Literacy
- **Technologies:** 3D Printing; SolidWorks; CAD; MATLAB/Simulink; ARM Development Studio; Microsoft Suite, C/C++, Python
- **Interests:** Custom PC Building; Automation; Orchestra; Baking

REFERENCES

- References available upon request.